

Advanced Circuits: Operational Amplifiers and Filters

A Reference Encyclopedia for Introduction to Electrical Engineering

Course Reference Document

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Abstract

This reference document distills the “Advanced Circuits” unit (operational amplifiers and filters) into a set of core concepts, each with expanded explanations, derivations, and worked examples. It is designed as a lookup resource: if you are stuck on a homework problem or a lab, find the relevant section and study the derivation and examples there. Python code is included where numerical verification builds insight; every script is self-contained and runs with only `numpy`, `scipy`, and `matplotlib`. Concepts covered: op-amp fundamentals and the open-loop model, negative feedback and the “golden rules,” the standard amplifier configurations, real op-amp limitations (bias currents, offsets, gain–bandwidth, slew rate), op-amp varieties (precision, programmable, single-supply, audio), instrumentation amplifiers, filter fundamentals and terminology, passive filters, and active filter design (Butterworth tables and Sallen–Key analysis).

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1 The Operational Amplifier

1.1 What an Op Amp Is

An **operational amplifier (op amp)** is a high-performance *differential* amplifier packaged as an integrated circuit. It has:

- a **noninverting input** V^+ ,
- an **inverting input** V^- ,
- an **output** terminal,
- two **DC supply pins** ($+V_S$ and $-V_S$), and
- occasionally specialized pins (e.g., offset null) used for fine-tuning.

Supply pins are usually omitted from schematics to reduce clutter. *If you see no supply leads on a schematic, assume a dual (split) supply is present.* The two supplies are commonly equal in magnitude (e.g., ± 15 V), but they need not be — single-supply operation (e.g., $+5$ V and ground) is covered in Section 4.3.

1.2 Open-Loop Behavior: The One Equation You Need

Everything about op-amp circuit analysis rests on a single expression relating the output to the two inputs through the **open-loop voltage gain** A_o :

The fundamental op-amp equation

$$V_{\text{out}} = A_o (V^+ - V^-) \quad (1)$$

The op amp amplifies only the *difference* between its inputs. For an ideal device $A_o \rightarrow \infty$; for real devices $A_o \approx 10^4$ to 10^6 .

Because A_o is enormous, even a microvolt-scale difference between V^+ and V^- drives the output to its limit. The output cannot exceed the supplies; it *saturates* at the **saturation voltages**, typically a volt or two inside each rail:

$$V_{\text{sat}}^+ \lesssim +V_S, \quad V_{\text{sat}}^- \gtrsim -V_S.$$

So, with no feedback:

- If $V^+ > V^-$: output slams to V_{sat}^+ (near $+V_S$).
- If $V^+ < V^-$: output slams to V_{sat}^- (near $-V_S$).

A bare op amp is therefore a **comparator**: a one-bit “which input is bigger?” detector.

1.2.1 The Ideal Op-Amp Rules

The three defining properties (“Rules”) of the op-amp model:

Rule 1: Gain. Ideal: $A_o = \infty$. Real: $A_o \approx 10^4$ – 10^6 .

Rule 2: Impedances. Ideal: input impedance $R_{\text{in}} = \infty$, output impedance $R_{\text{out}} = 0$. Real: $R_{\text{in}} \approx 10^6 \Omega$ (bipolar) to $10^{12} \Omega$ (JFET/MOSFET); $R_{\text{out}} \approx 10$ – 1000Ω .

Rule 3: Input current. The input terminals draw (essentially) no current: picoamps (FET) to nanoamps (bipolar) in real devices.

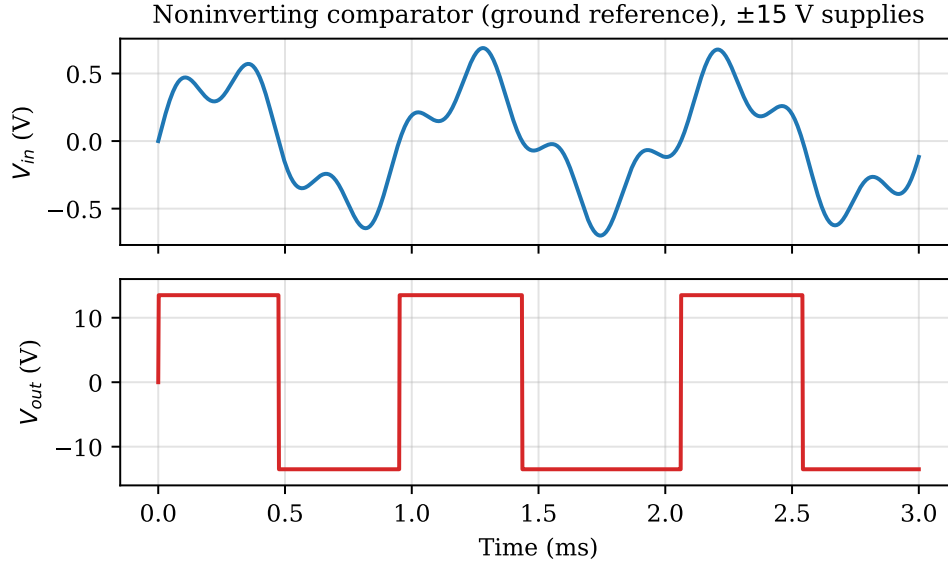


Figure 1: A noninverting comparator (V^- grounded). The output is a two-level signal indicating the sign of V_{in} ; the “gain” region is so narrow it is invisible at this scale.

Example 1.1: Comparator gains (from the notes, corrected and extended)

- (a) V^- grounded, V_{in} applied to V^+ . Then $V_{out} = A_o(V_{in} - 0) = A_o V_{in}$, so Gain = A_o (noninverting comparator).
- (b) V^+ grounded, V_{in} applied to V^- . Then $V_{out} = A_o(0 - V_{in}) = -A_o V_{in}$, so Gain = $-A_o$ (inverting comparator).
- (c) **Numerical extension.** Suppose $A_o = 2 \times 10^5$ and the supplies are ± 15 V with saturation at ± 13.5 V. What input range keeps configuration (a) *out* of saturation?

$$|V_{in}| < \frac{13.5 \text{ V}}{2 \times 10^5} = 67.5 \text{ } \mu\text{V}.$$

Any input outside a $\pm 67.5 \text{ } \mu\text{V}$ window saturates the output — this is why an op amp without feedback is only useful as a threshold detector.

1.3 The Water Analogy (Intuition Aid)

A useful mental model: pretend water *pressure* is voltage and water *flow* is current. The two inputs are tubes capped with elastic balloons (they take pressure but pass no flow — infinite input impedance). A lever between the balloons tilts according to which balloon is more inflated, rotating a valve that connects the output pipe either to a compressor (the $+V_S$ supply) or a vacuum (the $-V_S$ supply). The slightest pressure imbalance swings the valve fully one way — saturation. The output pipe delivers whatever flow the load demands without the pressure sagging (zero output impedance). Negative feedback, in this analogy, is a small return pipe from the output back to the inverting balloon: as the output pressure rises, it re-inflates the inverting balloon until the lever balances — and the system settles at exactly the pressure where the two balloons match.

1.4 What Is Inside (and Why You Should Not Worry About It)

Internally, a general-purpose bipolar op amp (like the classic 741) contains roughly twenty transistors organized in three stages:

1. a high-input-impedance **differential amplifier** (senses $V^+ - V^-$),
2. a **high-gain voltage amplifier** with a level shifter (provides most of A_o and lets the output swing both polarities), and
3. a **low-impedance output stage** (a push-pull buffer that drives the load).

Tracing currents through this network is unnecessary for circuit design. The entire point of the op-amp abstraction is that Rules 1–3 plus Eq. (1) predict terminal behavior without opening the box. This “cop-out” is not laziness — it is good engineering: design against the specified terminal behavior, not the internal implementation.

2 Negative Feedback and the Golden Rules

2.1 Why Feedback Tames the Op Amp

Feeding a fraction of the output back to the *inverting* input creates **negative feedback**: any rise in output opposes itself through the feedback path, so instead of saturating, the circuit settles at a stable equilibrium set by external components. The op amp’s absurd gain is deliberately “thrown away” in exchange for precision: the closed-loop gain depends (almost) only on resistor ratios, which can be accurate to 0.1% and stable over temperature — unlike A_o , which varies wildly between parts.

The Golden Rules (valid only with negative feedback)

1. **No input current**: the inputs draw no current (Rule 3 above).
 2. **Virtual short**: the op amp drives its output to whatever voltage makes $V^+ = V^-$.
- Golden Rule 2 is a *consequence* of infinite gain plus negative feedback, not an independent axiom — see the derivation below. It fails if the feedback is positive or the output saturates.

Derivation of Golden Rule 2 from finite gain

Let the feedback network return a fraction β of the output to the inverting input ($0 < \beta \leq 1$), with input signal V_{in} at the noninverting input:

$$V_{out} = A_o(V^+ - V^-) = A_o(V_{in} - \beta V_{out}) \implies V_{out} = \frac{A_o}{1 + A_o\beta} V_{in}.$$

The differential input voltage is then

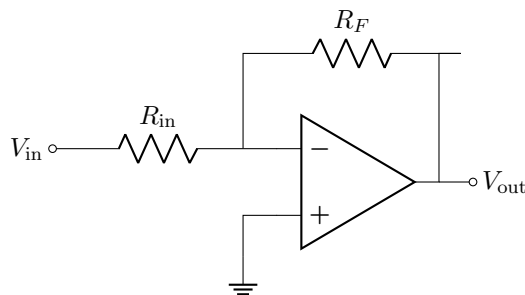
$$V^+ - V^- = \frac{V_{out}}{A_o} = \frac{V_{in}}{1 + A_o\beta} \xrightarrow{A_o \rightarrow \infty} 0.$$

With $A_o = 10^5$, $\beta = 0.1$, and $V_{in} = 1$ V, the input difference is a mere 100 μ V — for analysis purposes, zero. Hence: *the inputs sit at the same voltage*. Note also the closed-loop gain limit:

$$\frac{V_{out}}{V_{in}} = \frac{A_o}{1 + A_o\beta} \xrightarrow{A_o \rightarrow \infty} \frac{1}{\beta}, \quad (2)$$

which depends only on the feedback network. This single equation is the heart of every amplifier below.

2.2 The Inverting Amplifier



Analysis with the Golden Rules. V^+ is grounded, so by the virtual short $V^- = 0$ as well — a **virtual ground**. Since the inverting input draws no current, the current through R_{in} must continue through R_F :

$$i = \frac{V_{in} - 0}{R_{in}} = \frac{0 - V_{out}}{R_F} \implies \boxed{\frac{V_{out}}{V_{in}} = -\frac{R_F}{R_{in}}}$$

The minus sign means the output is inverted (180° phase shift for sinusoids). Increasing R_F increases the gain magnitude; decreasing it decreases the gain.

Two practical properties:

- **Input impedance** = R_{in} (the source looks into R_{in} terminated at a virtual ground). This can be undesirably low — a trade-off against the noninverting topology.
- **Gains below 1 are allowed** ($R_F < R_{in}$), unlike the noninverting amplifier.

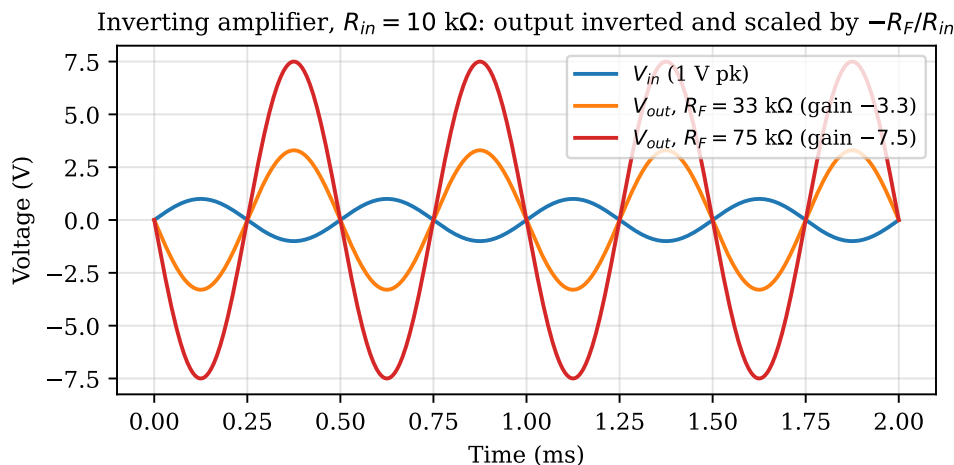


Figure 2: Inverting amplifier with $R_{in} = 10 \text{ k}\Omega$: doubling R_F (roughly) doubles the gain; the output is inverted.

Example 2.1: Finite-gain error in the inverting amplifier

Repeat the analysis *without* assuming $A_o = \infty$. KCL at the inverting node (voltage v^-),

with $V^+ = 0$ so $V_{\text{out}} = -A_o v^-$:

$$\frac{V_{\text{in}} - v^-}{R_{\text{in}}} = \frac{v^- - V_{\text{out}}}{R_F}, \quad v^- = -\frac{V_{\text{out}}}{A_o}.$$

Substituting and solving:

$$\frac{V_{\text{out}}}{V_{\text{in}}} = \frac{-R_F/R_{\text{in}}}{1 + \frac{1 + R_F/R_{\text{in}}}{A_o}}.$$

Numbers: $R_F = 100 \text{ k}\Omega$, $R_{\text{in}} = 10 \text{ k}\Omega$, $A_o = 2 \times 10^5$:

$$\frac{V_{\text{out}}}{V_{\text{in}}} = \frac{-10}{1 + 11/(2 \times 10^5)} = -9.99945,$$

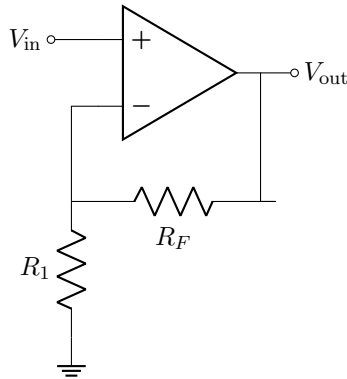
an error of 0.0055% from the ideal -10 — utterly negligible next to 1% resistor tolerances. This is *why* the ideal analysis is trusted.

Example 2.2: Design

Design an inverting amplifier with a gain of -4.7 and an input impedance of at least $20 \text{ k}\Omega$.

Solution. Input impedance $= R_{\text{in}}$, so choose $R_{\text{in}} = 20 \text{ k}\Omega$ (or larger). Then $R_F = 4.7 \times 20 \text{ k}\Omega = 94 \text{ k}\Omega$. Standard values: $R_{\text{in}} = 20 \text{ k}\Omega$, $R_F = 91 \text{ k}\Omega \Rightarrow$ gain -4.55 ; or use $R_{\text{in}} = 20 \text{ k}\Omega$ with $R_F = 94.2 \text{ k}\Omega$ ($91 \text{ k} + 3.2 \text{ k}$ in series) for -4.71 . Keep both resistors in the $\text{k}\Omega$ – $100 \text{ k}\Omega$ decade: too small wastes output current, too large invites noise and bias-current errors (Section 3.1).

2.3 The Noninverting Amplifier and the Follower



By the virtual short, $V^- = V_{\text{in}}$. The divider R_1 – R_F (unloaded, by Golden Rule 1) sets $V^- = V_{\text{out}} R_1 / (R_1 + R_F)$. Equating:

$$\frac{V_{\text{out}}}{V_{\text{in}}} = 1 + \frac{R_F}{R_1}$$

Here $\beta = R_1 / (R_1 + R_F)$, consistent with Eq. (2). Gain is always ≥ 1 ; input impedance is essentially the op amp's own (enormous) input impedance — the key advantage over the inverting topology.

Voltage follower. Set $R_F = 0$ (wire) and remove R_1 : gain $= 1$. Useless? No — it is an *impedance transformer*: near-infinite input impedance, near-zero output impedance. Use it to read

a high-impedance source (a sensor, a divider) without loading it.

Example 2.3: Buffering a sensor divider

A thermistor divider has a 10 k Ω Thévenin resistance and produces 2.500 V open-circuit. It drives a 47 k Ω load.

Unbuffered: $V_{\text{load}} = 2.500 \times \frac{47}{47+10} = 2.061 \text{ V}$ — a 17.6% error!

With a follower between divider and load: the follower input draws $\sim\text{pA}$, so $V_{\text{load}} = 2.500 \text{ V}$, and the follower's $\sim\text{m}\Omega$ -to- Ω -scale closed-loop output impedance drives 47 k Ω with negligible droop. The follower converts a fragile voltage into a stiff one.

2.4 Mathematical Circuits: Summing, Difference, Integrator, Differentiator

Negative feedback plus a change of components turns the op amp into an analog computer element. These four appear constantly in signal conditioning and are natural extensions of the notes.

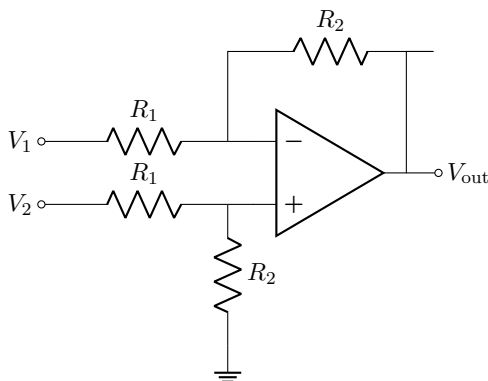
2.4.1 Summing amplifier

Multiple inputs $V_1, V_2, \dots$ through resistors $R_1, R_2, \dots$ into the virtual ground, with feedback R_F . Currents add at the node (KCL) and all flow through R_F :

$$V_{\text{out}} = -R_F \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \dots \right).$$

Equal resistors give a pure (inverted) sum; scaled resistors give a weighted sum — the basis of audio mixers and simple digital-to-analog converters.

2.4.2 Difference amplifier



With matched pairs as shown, superposition (or the Golden Rules directly) gives

$$V_{\text{out}} = \frac{R_2}{R_1} (V_2 - V_1).$$

This is the germ of the instrumentation amplifier (Section 5); its weakness is that the two inputs load the sources unequally and the resistors must be *very* well matched for good common-mode rejection.

2.4.3 Integrator and differentiator

Replace R_F with a capacitor C (input through R): the feedback current charges C , and

$$V_{\text{out}}(t) = -\frac{1}{RC} \int_0^t V_{\text{in}}(\tau) d\tau + V_{\text{out}}(0).$$

Swap the components (C in, R feedback) for the differentiator: $V_{\text{out}} = -RC dV_{\text{in}}/dt$. In practice the integrator needs a large resistor across C (to bleed off DC drift) and the differentiator needs a small resistor in series with C (to tame high-frequency noise) — both are first-order fixes you will meet in lab.

Example 2.4: Integrator turning a square wave into a triangle

$R = 10 \text{ k}\Omega$, $C = 10 \text{ nF}$, input a $\pm 1 \text{ V}$, 1 kHz square wave. During each half-period ($T/2 = 0.5 \text{ ms}$) the input is constant at $\pm 1 \text{ V}$, so the output ramps linearly with slope

$$\frac{dV_{\text{out}}}{dt} = -\frac{V_{\text{in}}}{RC} = \mp \frac{1 \text{ V}}{10^4 \times 10^{-8} \text{ s}} = \mp 10^4 \text{ V/s},$$

giving a triangle wave of peak-to-peak amplitude $10^4 \times 0.5 \times 10^{-3} = 5 \text{ V}$. The numerical simulation below (and Figure 3) confirms this.

```
1 import numpy as np, matplotlib.pyplot as plt
2 from scipy import signal
3 fs = 1e6
4 t = np.arange(0, 4e-3, 1/fs)
5 vin = signal.square(2*np.pi*1e3*t)           # +/-1 V, 1 kHz
6 R, C = 10e3, 10e-9
7 vout = -np.cumsum(vin)/fs/(R*C)              # numerical integral
8 plt.plot(t*1e3, vin); plt.plot(t*1e3, vout); plt.show()
9 print(vout.max() - vout.min())               # ~5.0 V peak-to-peak
```

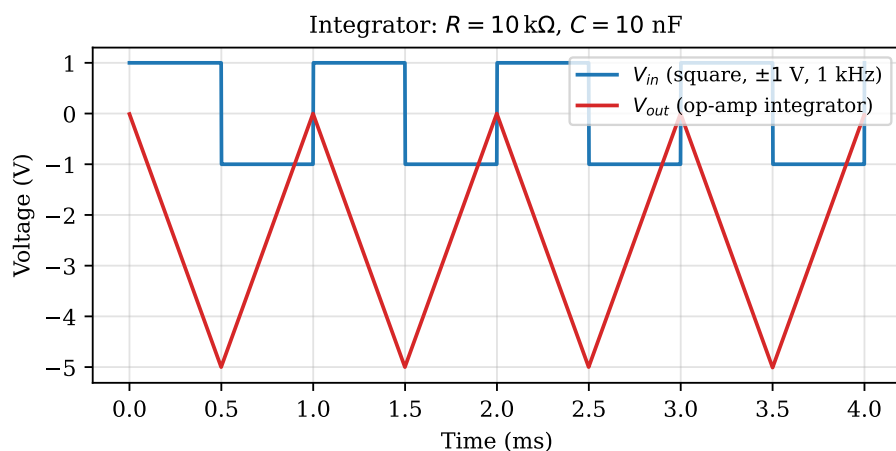


Figure 3: Op-amp integrator response to a square wave: linear ramps, i.e., a triangle wave of 5 V peak-to-peak, as predicted.

2.5 Positive Feedback and the Schmitt Trigger

Feedback to the *noninverting* input reinforces rather than opposes the output — the op amp is driven harder into saturation. This sounds useless but is exactly what a robust comparator needs. A plain comparator chatters when a noisy signal lingers near threshold; **positive feedback adds hysteresis**, creating a *Schmitt trigger* whose upper and lower thresholds differ.

With the input applied to V^- and a divider R_1/R_2 from output to V^+ (referenced to ground), the thresholds are

$$V_{T\pm} = \pm V_{\text{sat}} \frac{R_1}{R_1 + R_2},$$

so noise smaller than the hysteresis window $V_{T+} - V_{T-}$ cannot cause double-triggering. Schmitt triggers appear inside oscillators (relaxation oscillators), switch debouncers, and anywhere a clean logic edge must be extracted from a slow or noisy analog signal.

Common pitfall

The Golden Rules **do not apply** under positive feedback — there is no virtual short. Check the feedback path before invoking $V^+ = V^-$: feedback must return to the *inverting* input for the rules to hold.

3 Real Op Amps: Non-Idealities

Real devices deviate from the ideal rules in ways that matter once you design for precision, speed, or low power. The five deviations below cover most practical trouble.

3.1 Input Bias and Offset Currents

Real input transistors need a small DC current to operate: the **input bias current** I_B (nanoamps for bipolar inputs, picoamps or less for JFET/MOSFET inputs). This current flows through whatever resistances connect to each input, producing voltage drops that masquerade as signal and shift (*offset*) the output.

Compensation resistor. In an inverting amplifier, the inverting input sees a DC (Thévenin) resistance of $R_{\text{in}} \parallel R_F$, while the noninverting input sees $0\ \Omega$ (grounded directly). Placing a resistor

$$R_{\text{comp}} = R_{\text{in}} \parallel R_F = \frac{R_{\text{in}} R_F}{R_{\text{in}} + R_F}$$

between V^+ and ground makes both bias currents produce (nearly) equal drops, which the amplifier subtracts as common-mode. The residual error then depends only on the **input offset current** $I_{OS} = |I_{B+} - I_{B-}|$, typically 5–10× smaller than I_B itself.

Example 3.1: How big is the bias-current error?

Inverting amplifier, $R_{\text{in}} = 10\ \text{k}\Omega$, $R_F = 100\ \text{k}\Omega$, bipolar op amp with $I_B = 80\ \text{nA}$, $I_{OS} = 10\ \text{nA}$.

Without compensation: I_B into the inverting node flows through R_F (the node is a virtual ground), producing an output shift

$$\Delta V_{\text{out}} \approx I_B R_F = 80\ \text{nA} \times 100\ \text{k}\Omega = 8\ \text{mV}.$$

With $R_{\text{comp}} = 10\text{k} \parallel 100\text{k} = 9.09\ \text{k}\Omega$: the error scales down to $\Delta V_{\text{out}} \approx I_{OS} R_F = 1\ \text{mV}$.

With a JFET op amp ($I_B = 50$ pA): the uncompensated error is 5 μ V — compensation becomes unnecessary. Moral: match the input technology to the impedance level of your circuit.

3.2 Input Offset Voltage

Manufacturing mismatch between the input transistors means the output is zero not when $V^+ - V^- = 0$ but when it equals a small **input offset voltage** V_{OS} (bipolar: ~ 0.1 – 5 mV; precision parts: μ V). The amplifier processes V_{OS} like any input at the noninverting terminal, so it appears at the output multiplied by the *noninverting* gain ($1 + R_F/R_{in}$) — called the **noise gain** — regardless of the signal topology. Some op amps (e.g., the 741) provide offset-null pins for trimming with a potentiometer; precision and “zero-drift” op amps trim or chop it away internally.

Example 3.2: Offset at the output

A gain -100 inverting amplifier ($R_{in} = 1$ k Ω , $R_F = 100$ k Ω) uses an op amp with $V_{OS} = 2$ mV. Output error:

$$\Delta V_{out} = V_{OS} \left(1 + \frac{R_F}{R_{in}} \right) = 2 \text{ mV} \times 101 = 202 \text{ mV}.$$

If the signal of interest is a 10 mV sensor output (amplified to 1 V), a 202 mV offset is a 20% error — unacceptable. Fixes: trim it, AC-couple it, or specify a precision op amp with $V_{OS} \leq 25$ μ V (error ≤ 2.5 mV).

3.3 Gain–Bandwidth Product

The open-loop gain is huge only at DC. Internal compensation rolls A_o off at -20 dB/decade starting from a very low corner (often 1 – 10 Hz), so that

$$|A(jf)| \approx \frac{A_o}{\sqrt{1 + (f/f_c)^2}} \approx \frac{f_T}{f} \quad (f \gg f_c),$$

where $f_T = A_o f_c$ is the **unity-gain frequency**. The product of gain and frequency along this rolloff is constant: the **gain–bandwidth product** (GBW). Closing the loop at gain G buys you a closed-loop bandwidth

$$f_{3dB} \approx \frac{f_T}{G} \quad (3)$$

Gain and bandwidth trade one-for-one: see Figure 4.

Example 3.3: Staging for bandwidth

You need a gain of 1000 for a signal band extending to 10 kHz, using op amps with $f_T = 1$ MHz.

One stage: bandwidth $= 10^6/1000 = 1$ kHz. Fails.

Two stages of gain $\sqrt{1000} \approx 31.6$ each: each stage has bandwidth $10^6/31.6 = 31.6$ kHz. Two identical cascaded first-order stages shrink the composite bandwidth by $\sqrt{\sqrt{2} - 1} \approx 0.644$, giving ≈ 20 kHz overall. Passes with margin. Cascading moderate-gain stages is the standard cure for GBW limits.

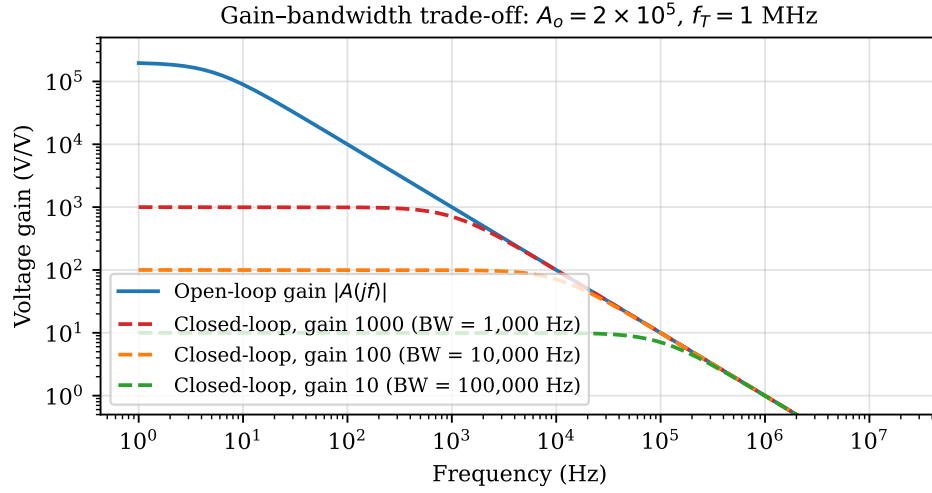


Figure 4: Open-loop gain rolloff and closed-loop responses for gains of 10, 100, 1000 with $f_T = 1$ MHz. Each factor of 10 in gain costs a factor of 10 in bandwidth.

3.4 Slew Rate

Independent of the small-signal GBW limit, the output voltage cannot change faster than the **slew rate** SR (in $V/\mu s$; e.g., $0.5 V/\mu s$ for the 741, 10–1000+ for fast parts). A sine output $V_p \sin(2\pi ft)$ has maximum slope $2\pi f V_p$, so distortion-free full-power operation requires

$$f \leq \frac{SR}{2\pi V_p}. \quad (4)$$

Example 3.4: Full-power bandwidth

A 741 ($SR = 0.5 V/\mu s$) must output a 10 V-peak sine. Maximum frequency:

$$f_{\max} = \frac{0.5 \times 10^6 \text{ V/s}}{2\pi \times 10 \text{ V}} \approx 8.0 \text{ kHz}.$$

Above 8 kHz the “sine” degenerates into a triangle. Note this is far below the 741’s small-signal GBW (~ 1 MHz): slew rate, not GBW, often sets the ceiling for *large* signals.

3.5 Input Technologies: Bipolar vs. JFET vs. MOSFET

Property	Bipolar	JFET	MOSFET
Input bias current	nA (high)	pA (low)	0.1 pA (lowest)
Input impedance	$\sim 10^6 \Omega$	$\sim 10^9\text{--}10^{12} \Omega$	$\sim 10^{12} \Omega$
Offset voltage	low	medium	medium–high
Offset drift	low	medium	medium
Bias matching	excellent	fair	fair
Bias vs. temperature	low variation	fair (doubles per $\sim 10^\circ\text{C}$)	fair

Phase inversion / latch-up (JFET hazard). If the common-mode input voltage of some JFET op amps approaches the negative supply too closely, the inputs can effectively swap roles — negative feedback becomes positive — and the output latches to a rail. Avoid by restricting the common-mode range or choosing a bipolar or rail-to-rail part.

Reading a datasheet. Rather than memorizing technology folklore, compare the specifications directly: slew rate, noise, V_{OS} and drift, I_B and drift, common-mode range, gain, bandwidth, input/output impedance, supply limits, supply current, power dissipation, temperature range, and whether the part is internally or externally frequency-compensated (externally compensated parts need added components to remain stable but can be faster).

4 Op Amp Varieties

4.1 General-Purpose and Precision

General-purpose parts (741-class, TL08x, LF412) balance cost against moderate specs. **Precision** op amps (OP-27, LT1057, AD713 class) are engineered for low V_{OS} , low drift, low bias current, and low noise — for instrumentation front-ends where microvolts matter.

4.2 Programmable Op Amps: Then and Now

The classic approach (LM4250). A “programmable” op amp of the 1970s–80s is one whose internal operating currents are set by an external **programming current** I_{set} , drawn from a programming pin through a resistor to ground. Quiescent power, bias currents, slew rate, GBW, and noise all scale roughly proportionally with I_{set} (from microamps to milliamps). One device can thus masquerade as many different op amps, and at tiny I_{set} it runs on very low supply voltages (~ 1 V) and microwatt power — ideal for battery operation. The trade: everything slows down together (low I_{set} means low slew rate and low GBW).

The contemporary approach (digitally programmable). Modern parts replace the analog I_{set} pin with a digital interface. Today “programmable” usually means a **programmable-gain amplifier (PGA)**: an op amp with an internal, factory-trimmed feedback network whose gain (and often input channel) is selected over SPI or I²C by a microcontroller. Two representative current-production families:

- **Texas Instruments PGA112/PGA113:** zero-drift input (~ 100 μ V offset), rail-to-rail output, SPI-selected gains (binary steps 1–128 or “scope” steps 1–200), built-in input multiplexer and calibration channels.
- **Microchip MCP6S91/21 family:** single-supply SPI PGAs with rail-to-rail input/output and gains of 1–32, plus multiplexed inputs and software shutdown.

The philosophical shift: the LM4250 let you trade *speed for power* with a resistor; a PGA lets firmware retune *gain and routing* on the fly — e.g., auto-ranging a sensor front-end so a single ADC input handles both microvolt and volt-scale signals. Both embody the same idea — one device, many personalities — with the “knob” moved from the analog domain to the digital one. (Low-power operation, meanwhile, migrated into dedicated nanowatt op-amp families rather than a programming pin.)

4.3 Single-Supply Op Amps

Designed to run from one positive supply (e.g., +5 or +12 V) with a common-mode input range extending down to the negative rail (ground). **The output cannot go below ground**, so a bipolar

(AC) signal must be re-centered before amplification: bias the noninverting input at mid-supply with a divider, AC-couple the input, and the output then swings around $V_S/2$.

Example 4.1: Mid-rail biasing

A +5 V single-supply follower must pass a ± 1 V audio signal. Bias V^+ with two 100 k Ω resistors from +5 V to ground ($V_{DC}^+ = 2.5$ V) and couple the signal in through a capacitor. The output swings 2.5 ± 1 V = 1.5–3.5 V — comfortably inside the rails. Choose the coupling capacitor so the high-pass corner $f_c = 1/(2\pi R_{\text{bias,eq}} C)$ sits below the signal band, with $R_{\text{bias,eq}} = 100\text{k} \parallel 100\text{k} = 50$ k Ω : $C = 1$ μF gives $f_c = 3.2$ Hz.

4.4 Audio Amplifiers

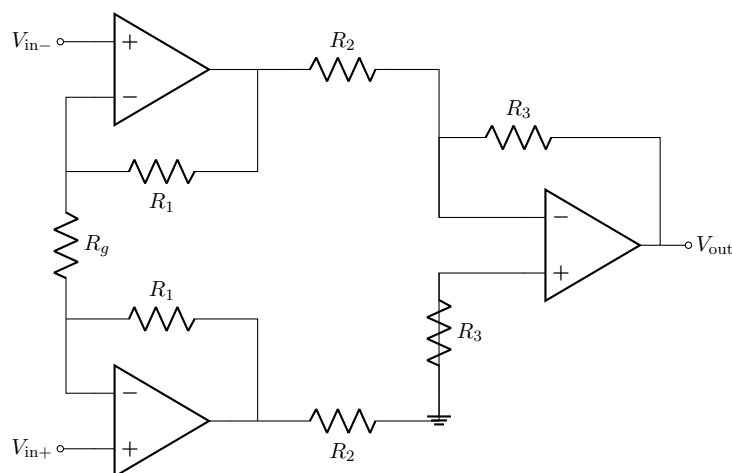
Audio power amplifiers (LM386, LM383) are op-amp relatives optimized for the 20 Hz–20 kHz band, low crossover distortion, and driving low-impedance loads directly. The LM386 has an internally fixed gain of 20, raisable to 200 with an external RC across its gain pins (1 and 8), drives 8 Ω speakers, and runs from a single +4 to +12 V supply — a staple of battery-powered audio. The LM383 is a 3.5 A power device for 4 Ω loads with thermal shutdown. The design habits are the same as for op amps; only the output muscle differs.

5 Instrumentation Amplifiers

5.1 Why the Difference Amplifier Is Not Enough

The one-op-amp difference amplifier (Section 2.4) amplifies $V_2 - V_1$, but (i) its inputs load the sources through R_1 — badly, and unequally, if the sources have significant impedance (electrodes on skin can be 10–100 k Ω !) — and (ii) its **common-mode rejection** depends entirely on exquisite resistor matching. The **instrumentation amplifier (in-amp)** fixes both with buffered inputs.

5.2 The Three-Op-Amp Architecture



Derivation of the gain. The two input op amps have negative feedback, so the virtual short pins their inverting inputs at $V_{\text{in-}}$ and $V_{\text{in+}}$ respectively. The full differential input therefore appears

across R_g :

$$i_g = \frac{V_{in+} - V_{in-}}{R_g}.$$

No current enters the op-amp inputs, so i_g flows through both R_1 resistors. The voltage between the two buffer outputs is

$$V_{o2} - V_{o1} = i_g (R_1 + R_g + R_1) = (V_{in+} - V_{in-}) \left(1 + \frac{2R_1}{R_g}\right).$$

The first stage thus amplifies the *differential* signal by $(1 + 2R_1/R_g)$ while passing *common-mode* voltage at unity gain (if both inputs move together, no current flows in R_g and both outputs simply follow). The output difference amplifier contributes R_3/R_2 and rejects the common mode:

$$\boxed{\frac{V_{out}}{V_{in+} - V_{in-}} = \left(1 + \frac{2R_1}{R_g}\right) \frac{R_3}{R_2}} \quad (5)$$

If R_g is omitted (open-circuited), the first stage has unity differential gain. A single resistor R_g sets the entire gain — which is why commercial in-amps (with laser-trimmed internal R_1, R_2, R_3) bring out only the two R_g pins. The R_1 pair must be matched, and the output-stage resistor ratios even more so; integrated matching is why an in-amp IC beats three discrete op amps.

Figure of merit: CMRR. The common-mode rejection ratio compares differential gain to common-mode gain, in dB: $\text{CMRR} = 20 \log_{10}(A_d/A_{cm})$. Good in-amps exceed 100 dB.

Example 5.1: ECG front-end (biomedical application)

An ECG electrode pair delivers a 1 mV differential heartbeat signal riding on roughly 1.5 V of electrode half-cell potential mismatch and mains-induced common-mode pickup of 20 mV at 60 Hz. Design an in-amp stage with differential gain 1000 using the topology above with $R_3/R_2 = 10$ and $R_1 = 25 \text{ k}\Omega$.

First stage gain needed: $1000/10 = 100 = 1 + 2R_1/R_g \Rightarrow R_g = \frac{2 \times 25\text{k}}{99} = 505 \text{ }\Omega$ (choose 510 Ω , gain ≈ 990).

Common-mode output with CMRR = 100 dB: $A_{cm} = A_d/10^5 = 990/10^5 \approx 0.0099$, so the 20 mV of 60 Hz pickup contributes $0.0099 \times 20 \text{ mV} \approx 0.2 \text{ mV}$ at the output, versus the amplified heartbeat at $\approx 1 \text{ V}$. The interference is 74 dB below the signal — workable; a notch filter (Section 6) cleans up the rest.

6 Filters: Fundamentals

6.1 The Four Filter Types

A **filter** passes a chosen range of frequencies and blocks others:

- **Low-pass:** passes below a cutoff, blocks above (anti-alias filters, smoothing).
- **High-pass:** passes above a cutoff (AC coupling, blocking 60 Hz hum below audio, electrode drift removal).
- **Bandpass:** passes a band around a center frequency (radio tuning, isolating a heart-rate band).
- **Notch (band-reject):** blocks a narrow band, passes everything else (60 Hz mains rejection in ECG).

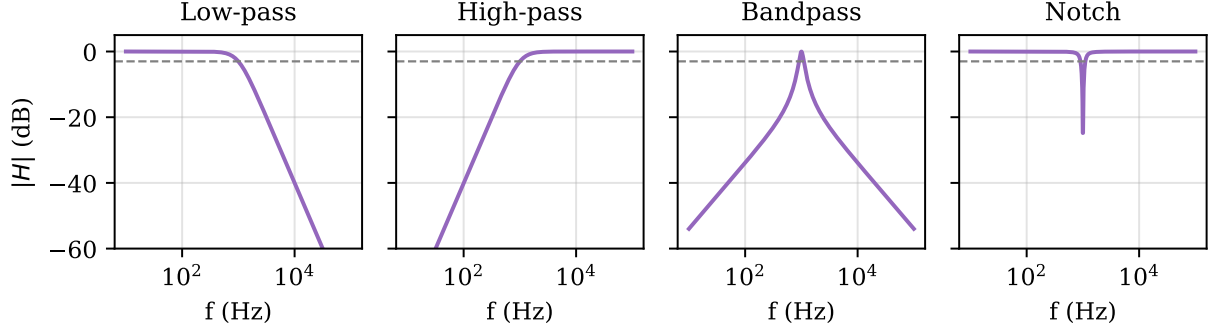


Figure 5: Magnitude responses of the four filter types (second-order examples, $f_0 = 1$ kHz). Dashed line: the -3 dB level.

Applications are everywhere: power supplies (smoothing rectified DC, killing line noise), radio (transmit and receive selectivity), audio (crossover networks routing lows to woofers and highs to tweeters), and biomedical instrumentation (isolating physiological bands, rejecting mains interference).

6.2 Terminology You Must Know

Filter behavior is described by a **response curve**: attenuation $|V_{\text{out}}/V_{\text{in}}|$ versus frequency, usually with attenuation in decibels,

$$|H|_{\text{dB}} = 20 \log_{10} \left| \frac{V_{\text{out}}}{V_{\text{in}}} \right|,$$

and frequency either as f (Hz) or $\omega = 2\pi f$ (rad/s).

3-dB frequency $f_{3\text{dB}}$ (cutoff).

Where the output has dropped 3 dB below the passband level: voltage down by $1/\sqrt{2} \approx 0.707$, power down by exactly half. Low- and high-pass filters have one; bandpass and notch filters have two, f_1 and f_2 .

Center frequency f_0 .

For a bandpass on a log-frequency axis the response is geometrically symmetric about

$$f_0 = \sqrt{f_1 f_2},$$

the *geometric* mean. For narrow-band filters ($f_2/f_1 < 1.1$) the arithmetic mean $f_0 \approx (f_1 + f_2)/2$ is an acceptable approximation.

Passband.

Frequencies reaching the output with at most 3 dB attenuation.

Stop-band frequency f_s .

The frequency at which attenuation first reaches a designer-specified value (e.g., 60 dB); beyond it lies the *stop band*.

Quality factor Q .

For a bandpass filter,

$$Q = \frac{f_0}{f_2 - f_1},$$

the center frequency divided by the 3-dB bandwidth: high Q = narrow, selective response. The same definition is conventionally used for the notch filter (high Q = narrow notch). *Note:* the source notes invert this ratio for the notch case; we adopt the standard convention here, and simply remember that a notch is often specified by its notch *width* $f_2 - f_1 = f_0/Q$ directly.

Order n and rolloff.

An n th-order filter ultimately rolls off at $20n$ dB/decade = $6n$ dB/octave. (Decade = factor of 10 in frequency; octave = factor of 2; 20 dB/decade ≈ 6.02 dB/octave.)

Example 6.1: Bandpass bookkeeping

A bandpass filter has 3-dB points at $f_1 = 900$ Hz and $f_2 = 1130$ Hz.

Center: $f_0 = \sqrt{900 \times 1130} = 1008$ Hz. (Arithmetic estimate: 1015 Hz — close, since $f_2/f_1 = 1.26$ is only moderately wide.)

Bandwidth: $f_2 - f_1 = 230$ Hz. $Q = 1008/230 = 4.4$.

A Q of 4.4 is “moderately selective”: the filter passes a band about 23% as wide as its center frequency.

6.3 The First-Order RC Low-Pass: Full Derivation

The simplest filter: series R , shunt C , output across C . Treat it as a frequency-dependent voltage divider with capacitor impedance $Z_C = 1/(j\omega C)$:

$$H(j\omega) = \frac{V_{\text{out}}}{V_{\text{in}}} = \frac{Z_C}{R + Z_C} = \frac{1}{1 + j\omega RC}.$$

Magnitude and phase:

$$|H| = \frac{1}{\sqrt{1 + (\omega RC)^2}}, \quad \angle H = -\arctan(\omega RC).$$

Setting $|H| = 1/\sqrt{2}$ gives $\omega_c RC = 1$:

$$f_{3\text{dB}} = \frac{1}{2\pi RC}$$

Well above cutoff, $|H| \approx 1/(\omega RC)$: halving per octave, i.e., -6 dB/octave (-20 dB/decade) — the signature first-order rolloff. The high-pass twin (swap R and C) gives $H = j\omega RC/(1 + j\omega RC)$ with the same cutoff. The inductor versions ($f_c = R/2\pi L$) behave analogously but are avoided when possible (Section 6.4).

Example 6.2: Attenuation at a given frequency

An RC low-pass with $R = 1.6$ k Ω , $C = 100$ nF: $f_{3\text{dB}} = 1/(2\pi \times 1600 \times 10^{-7}) \approx 1$ kHz. Attenuation at 10 kHz?

$$|H| = \frac{1}{\sqrt{1 + (10)^2}} = \frac{1}{\sqrt{101}} = 0.0995 \Rightarrow 20 \log_{10}(0.0995) = -20.0 \text{ dB}.$$

One decade above cutoff $\rightarrow 20$ dB down, as the rolloff rule predicts. At 40 kHz (two octaves further): -32 dB.

6.4 Passive Filters and Their Limits

Passive filters (R, L, C only) work best from roughly 100 Hz to 300 MHz:

- *Below* ~100 Hz, required L and C values become physically enormous;
- *Above* a few hundred MHz, parasitic capacitance and inductance of the components themselves take over.

They add no noise, need no power, and handle RF effortlessly — at radio frequencies passive is the default. Their weaknesses: no gain (only loss), sensitivity to source and load impedances, and awkward cascading.

Why cascading passive sections is not free. Stacking identical RC sections to steepen the rolloff fails to deliver the naive expectation, because each section *loads* the previous one (the second section’s input impedance is not infinite). For two identical RC low-pass sections connected directly, nodal analysis gives

$$H(s) = \frac{1}{1 + 3sRC + (sRC)^2} \quad (s = j\omega),$$

not the buffered product $1/(1 + sRC)^2$ — note the “3” where a “2” would otherwise stand. The result: more passband droop and a lower, mushier corner (Figure 6). Beyond loading, multi-section passive design must contend with transient response, phase shift, component losses, winding capacitance, and magnetic pickup in inductors. Active filters sidestep all of this.

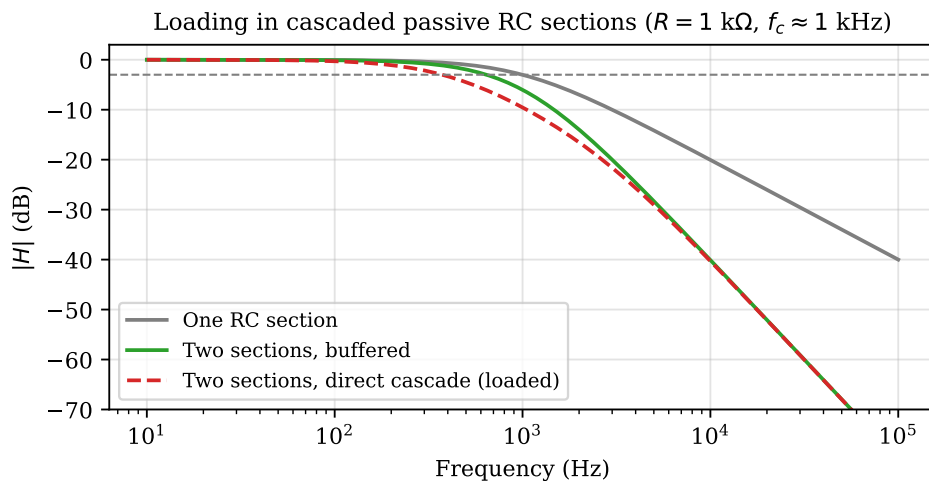


Figure 6: Two cascaded RC sections: direct connection (loaded) vs. separated by a buffer. Loading drags the -3 dB corner from 644 Hz down to 374 Hz and softens the knee.

6.5 Active Filters: Pros and Cons

Active filters combine op amps with only R’s and C’s — no inductors:

- + Work down to (nearly) 0 Hz; no giant inductors.
- + Can provide gain, not just loss.
- + Op-amp outputs isolate sections: cascade freely, no loading, and input/output impedances are set by design, independent of frequency.
- + Match LC-filter performance while being easier and cheaper to build.

- Limited high-frequency range: above ~ 100 kHz (op-amp GBW and slew-rate limits, Section 3.3) they grow unreliable. *At radio frequencies, use passive filters.*
- Need power supplies; add (some) noise.

7 Active Filter Design

7.1 The Butterworth Response

Simple cascaded identical sections give a droopy passband. The classical filter responses fix this by choosing pole positions deliberately. The three you will meet:

- **Butterworth** — “maximally flat” passband; the standard general-purpose choice, and the one used in this course.
- **Chebyshev** — steeper initial rolloff bought with passband ripple.
- **Bessel** — best (most linear) phase response, gentlest rolloff; preserves pulse shapes.

The Butterworth low-pass magnitude, normalized to a 1 rad/s cutoff, is

$$|H(j\omega)| = \frac{1}{\sqrt{1 + \omega^{2n}}}, \quad (6)$$

which is -3 dB at $\omega = 1$ for *every* order n , flat as possible at DC, and $-20n$ dB/decade asymptotically.

Choosing the order analytically. Requiring attenuation A_s (in dB) at normalized stop frequency ω_s :

$$n \geq \frac{\log_{10}(10^{A_s/10} - 1)}{2 \log_{10}(\omega_s)} \quad (7)$$

This replaces reading the response-curve graphs (Figure 7) with one line of arithmetic.

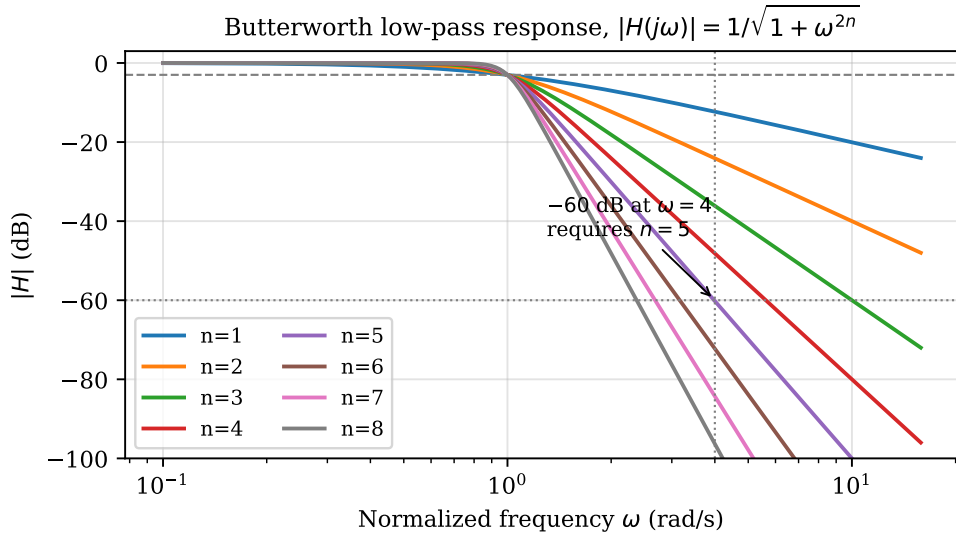


Figure 7: Normalized Butterworth responses, orders 1–8. To get 60 dB of attenuation by $\omega = 4$, the $n = 5$ curve is the first that suffices — matching Eq. (7).

7.2 The Normalization–Scaling Workflow

All table-based active filter design follows four steps:

1. **Normalize.** Compute the *steepness factor* $A_s = f_s/f_{3\text{dB}}$ (low-pass) — the stop frequency in units of the cutoff.
2. **Pick the order** from Eq. (7) or the curves.
3. **Build the normalized filter** ($\omega_c = 1 \text{ rad/s}$, $R = 1 \Omega$) by cascading the tabulated two-pole and three-pole unity-gain sections (Table 1).
4. **Scale** to the real cutoff and practical impedances:

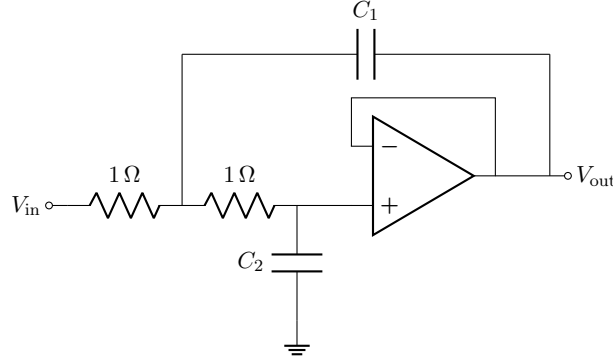
$$\boxed{C_{\text{actual}} = \frac{C_{\text{table}}}{Z \cdot 2\pi f_{3\text{dB}}}, \quad R_{\text{actual}} = Z \cdot R_{\text{table}}} \quad (8)$$

where Z (typically 10^4) is an impedance-scale factor chosen to land components in practical decades. Only capacitors are frequency-scaled (resistors are not reactive); both are impedance-scaled, in opposite directions, which leaves the frequency response untouched.

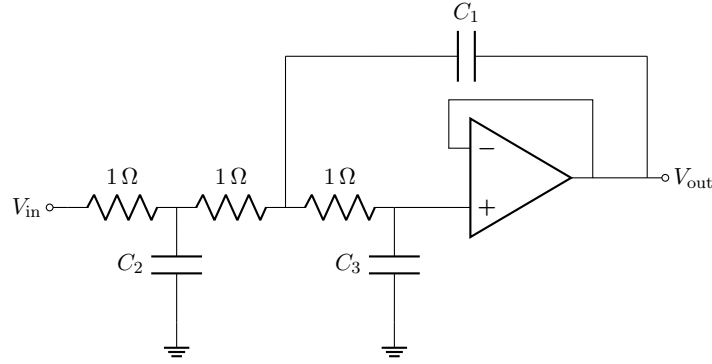
Table 1: Butterworth normalized active low-pass filter values (unity-gain sections; capacitors in farads for $\omega_c = 1 \text{ rad/s}$, $R = 1 \Omega$).

Order n	Sections	Type	C_1	C_2	C_3
2	1	2-pole	1.414	0.7071	—
3	1	3-pole	3.546	1.392	0.2024
4	2	2-pole	1.082	0.9241	—
		2-pole	2.613	0.3825	—
5	2	3-pole	1.753	1.354	0.4214
		2-pole	3.235	0.3090	—
6	3	2-pole	1.035	0.9660	—
		2-pole	1.414	0.7071	—
		2-pole	3.863	0.2588	—
7	3	3-pole	1.531	1.336	0.4885
		2-pole	1.604	0.6235	—
		2-pole	4.493	0.2225	—
8	4	2-pole	1.020	0.9809	—
		2-pole	1.202	0.8313	—
		2-pole	2.000	0.5557	—
		2-pole	5.758	0.1950	—

The basic building blocks are the unity-gain **two-pole (Sallen–Key) section** and **three-pole section**:



Two-pole (Sallen-Key) section



Three-pole section

Example 7.1: Fifth-order low-pass, $f_{3\text{dB}} = 100 \text{ Hz}$, $\geq 60 \text{ dB}$ at 400 Hz

Step 1. $A_s = 400/100 = 4$: the stop frequency sits at $\omega = 4$ on the normalized axis.

Step 2. Eq. (7): $n \geq \log_{10}(10^6 - 1)/(2 \log_{10} 4) = 6.0/1.204 = 4.98 \Rightarrow n = 5$.

Step 3. From Table 1, $n = 5$: a three-pole section ($C_1 = 1.753$, $C_2 = 1.354$, $C_3 = 0.4214$) cascaded with a two-pole section ($C_1 = 3.235$, $C_2 = 0.3090$).

Step 4. Scale with $Z = 10^4$ and $2\pi f_{3\text{dB}} = 628.3 \text{ rad/s}$, so each $C_{\text{actual}} = C_{\text{table}}/(6.283 \times 10^6)$:

Component	Table value	Actual value
3-pole C_1	1.753 F	0.279 μF
3-pole C_2	1.354 F	0.216 μF
3-pole C_3	0.4214 F	0.0671 μF
2-pole C_1	3.235 F	0.515 μF
2-pole C_2	0.3090 F	0.0492 μF
All resistors	1 Ω	10 k Ω

(The source notes round these to 0.28, 0.22, 0.07, 0.52, 0.05 μF .) Figure 8 verifies the design numerically: -3.0 dB at 100 Hz and -60.2 dB at 400 Hz — the spec is met with almost no margin, exactly as the order calculation ($n = 4.98$) hinted.

```

1 import numpy as np
2 from scipy import signal
3 b, a = signal.butter(5, 2*np.pi*100, 'low', analog=True)
4 for f in (100, 400):

```

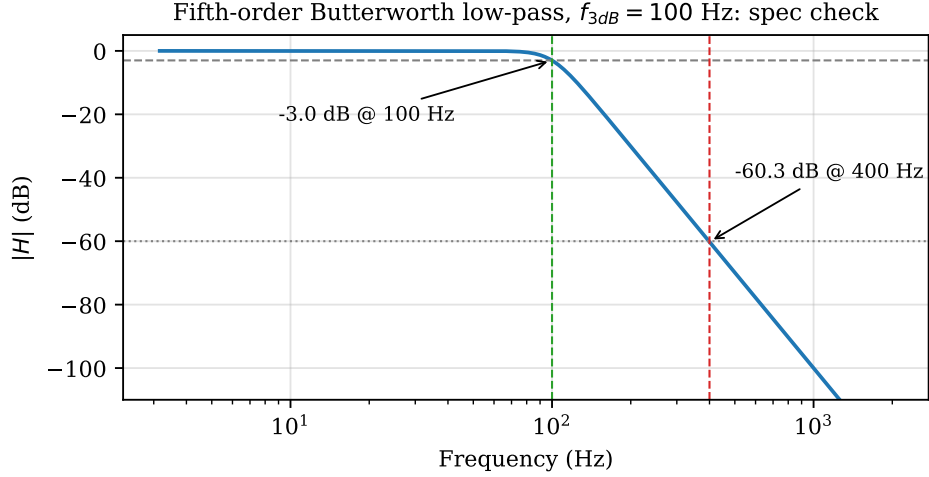


Figure 8: Numerical check of Example 7.1 via `scipy.signal.butter`.

```

5     _, h = signal.freqs(b, a, [2*np.pi*f])
6     print(f, 20*np.log10(abs(h[0])))
7     # 100 -3.01 dB | 400 -60.21 dB -> spec met

```

7.3 Second-Order Sallen–Key: The Math Behind the Table

Where do the table’s capacitor values come from? Analyze the unity-gain, equal- R Sallen–Key low-pass directly. Label the node between the two resistors v_x ; the op amp is a follower, so its output equals $V^+ = v_o$, and C_1 connects v_x to the *output* (bootstrapped feedback). KCL at the two nodes (in the Laplace domain, $s = j\omega$):

$$\frac{V_{in} - v_x}{R} = \frac{v_x - v_o}{1/(sC_1)} + \frac{v_x - v_o}{R}, \quad \frac{v_x - v_o}{R} = \frac{v_o - 0}{1/(sC_2)} \cdot 1 \text{ (current into } C_2\text{)}.$$

Solving the second equation for $v_x = v_o(1 + sRC_2)$ and substituting into the first yields, after collecting terms,

$$H(s) = \frac{v_o}{V_{in}} = \frac{1}{s^2 R^2 C_1 C_2 + 2sRC_2 + 1} \quad (9)$$

Comparing with the standard second-order low-pass form

$$H(s) = \frac{\omega_0^2}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2}$$

identifies

$$\omega_0 = \frac{1}{R\sqrt{C_1 C_2}}, \quad Q = \frac{1}{2} \sqrt{\frac{C_1}{C_2}} \quad (10)$$

Two clean insights:

- ω_0 depends on the *geometric mean* of the capacitors; Q depends only on their *ratio*. You can set them independently.

• **Consistency check with Table 1:** the $n = 2$ entry has $C_1 = 1.414 = \sqrt{2}$ and $C_2 = 0.7071 = 1/\sqrt{2}$. Then $\omega_0 = 1/\sqrt{\sqrt{2} \cdot 1/\sqrt{2}} = 1 \text{ rad/s}$ ✓ and $Q = \frac{1}{2}\sqrt{\sqrt{2}/(1/\sqrt{2})} = \frac{1}{2}\sqrt{2} = 0.7071$ — precisely the Butterworth Q . The table is just Eq. (10) solved for the classical pole positions. Figure 9 shows why Q matters: $Q < 0.707$ droops early; $Q > 0.707$ peaks before rolling off; $Q = 0.707$ is the flattest possible — Butterworth.

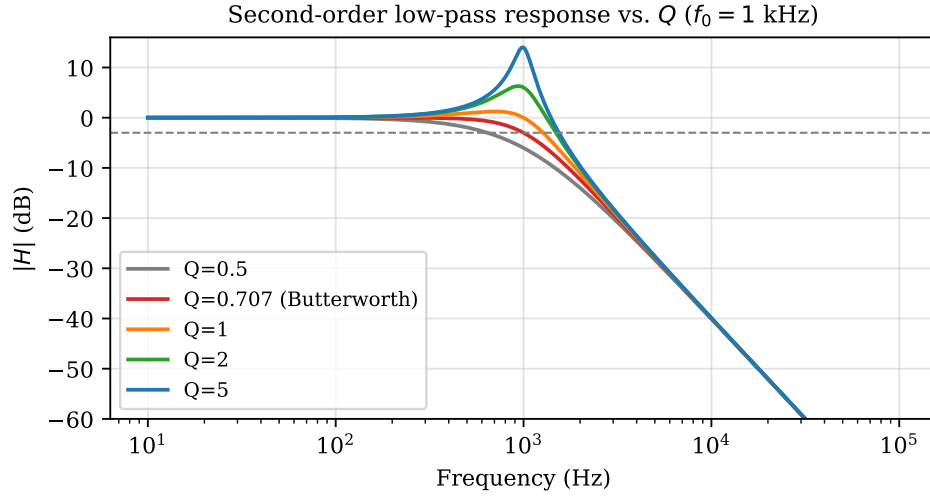


Figure 9: Effect of Q on a second-order low-pass. Butterworth ($Q = 0.7071$) is the boundary between droop and peaking.

Example 7.2: Direct Sallen–Key design at 2 kHz

Design a second-order Butterworth low-pass with $f_0 = 2 \text{ kHz}$ using equal resistors $R = 10 \text{ k}\Omega$.

From Eq. (10) with $Q = 1/\sqrt{2}$: $C_1/C_2 = 4Q^2 = 2$, and $\sqrt{C_1 C_2} = 1/(\omega_0 R) = 1/(2\pi \cdot 2000 \cdot 10^4) = 7.958 \text{ nF}$. Hence

$$C_1 = \sqrt{2} \times 7.958 \text{ nF} = 11.25 \text{ nF}, \quad C_2 = 7.958/\sqrt{2} = 5.63 \text{ nF}.$$

Practical values: $C_1 = 11 \text{ nF}$ ($10 \text{ nF} \parallel 1 \text{ nF}$), $C_2 = 5.6 \text{ nF}$. Verification (Figure 10): -3.01 dB at 2 kHz and -40 dB at 20 kHz (one decade $\rightarrow 20 \times 2 = 40 \text{ dB}$ for $n = 2$). *Sanity check against the table route:* scaling the table's $\sqrt{2}$ and $1/\sqrt{2} \text{ F}$ by $Z = 10^4$ and $2\pi \cdot 2000$ gives the same $11.25/5.63 \text{ nF}$ — the two methods are one method.

7.4 Active High-Pass Design by Transformation

High-pass design reuses the low-pass machinery via a normalized transformation:

1. **Translate the spec.** The high-pass steepness factor is $A_s = f_{3\text{dB}}/f_s$ (note: inverted relative to low-pass, since the stop band now lies *below* cutoff).
2. **Design the normalized low-pass** for that A_s (order from Eq. (7), sections from Table 1).
3. **Transform LP $\rightarrow$ HP:** replace every resistor R with a capacitor of value $\frac{1}{R} \text{ F}$, and every capacitor C with a resistor of value $\frac{1}{C} \Omega$. (This is the substitution $s \rightarrow 1/s$, which maps

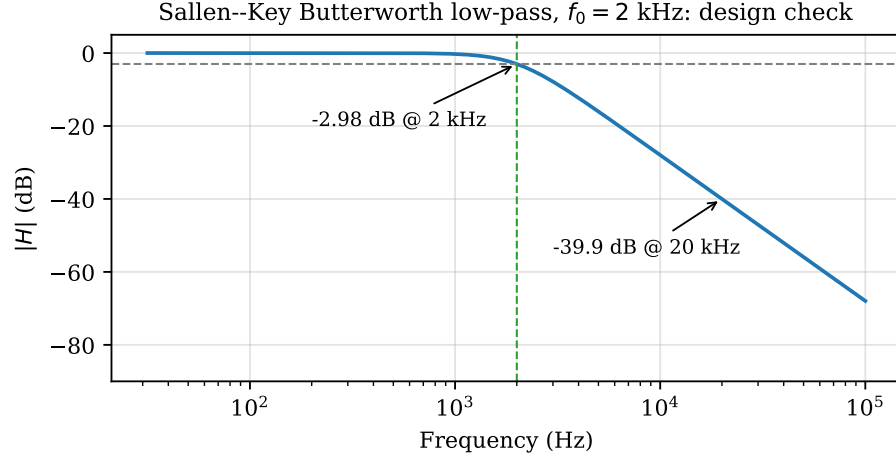


Figure 10: Design check for Example 7.2.

$\omega \rightarrow 1/\omega$ and mirrors the response about $\omega = 1$.)

4. **Scale** with the same rules, Eq. (8).

Example 7.3: High-pass, $f_{3dB} = 1$ kHz, ≥ 50 dB at 300 Hz

Step 1. $A_s = 1000/300 = 3.33$.

Step 2. $n \geq \log_{10}(10^5 - 1)/(2 \log_{10} 3.33) = 5.0/1.046 = 4.78 \Rightarrow n = 5$: three-pole + two-pole cascade.

Step 3. Transform the $n = 5$ normalized low-pass. Every 1Ω resistor becomes a 1 F capacitor; the tabulated capacitors become resistors:

LP element (table)	HP element (transformed)	After scaling ($Z = 10^4$, $f_{3dB} = 1$ kHz)
3-pole $C_1 = 1.753$ F	$R_1 = 1/1.753 = 0.570 \Omega$	5.70 k Ω
3-pole $C_2 = 1.354$ F	$R_2 = 0.739 \Omega$	7.39 k Ω
3-pole $C_3 = 0.4214$ F	$R_3 = 2.373 \Omega$	23.7 k Ω
2-pole $C_1 = 3.235$ F	$R_1 = 0.309 \Omega$	3.09 k Ω
2-pole $C_2 = 0.3090$ F	$R_2 = 3.236 \Omega$	32.4 k Ω
each $R = 1 \Omega$	$C = 1$ F	$C = \frac{1}{10^4 \cdot 2\pi \cdot 1000} = 15.9$ nF

(The notes round to 5.7 k, 7.4 k, 23.7 k, 3.1 k, 32.4 k, and $0.016 \mu\text{F}$.) Figure 11 confirms: -3 dB at 1 kHz, -52.3 dB at 300 Hz — spec met with 2 dB to spare, consistent with rounding n up from 4.78.

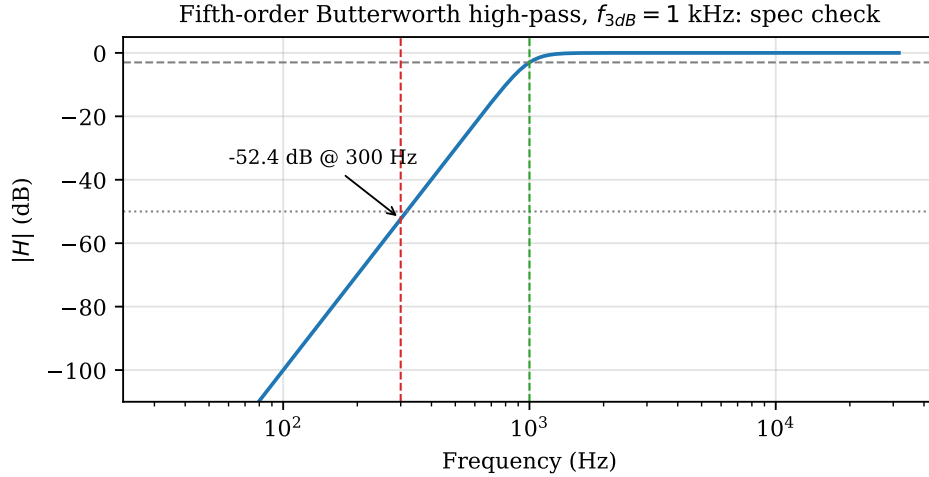


Figure 11: Numerical check of the high-pass design of Example 7.3.

7.5 Design Checklist and Op-Amp Requirements

Active filter design checklist

1. Spec: $f_{3\text{dB}}$, stop frequency f_s , required stop-band attenuation A_s (dB).
2. Steepness factor: LP $A_s^{(\omega)} = f_s/f_{3\text{dB}}$; HP $A_s^{(\omega)} = f_{3\text{dB}}/f_s$.
3. Order from Eq. (7); sections from Table 1. (HP: transform elements.)
4. Scale via Eq. (8); pick Z so capacitors land in the nF– μ F decade.
5. Check the op amp: GBW comfortably above $\sim 100\times$ the highest f_0 times the section gain, and slew rate adequate for the largest, fastest signals (Section 3.3).
6. Verify numerically (`scipy.signal.butter` + `freqs`) before building.

8 Corrections to the Source Notes

For instructors and careful readers, discrepancies found in the source document and resolved here:

1. **Notch-filter Q :** the notes give $Q = (f_2 - f_1)/f_0$ for the notch; the standard convention $Q = f_0/(f_2 - f_1)$ (high Q = narrow notch) is used throughout this document.
2. **Voltage at -3 dB:** stated as reduced “by $1/\sqrt{2}$ ”; precisely, the voltage is reduced *to* $1/\sqrt{2} \approx 70.7\%$ of its passband value (a reduction *by* 29.3%).
3. **Cross-references:** many figure numbers in the source are broken or missing captions (e.g., “see Figure ” with no number; “Figure 197 shows a simple DC amplifier” should point to the single-supply figure, 198); all references here are internally consistent.
4. **Filter handbook author:** “Zverck” should read *Zverev* (A. I. Zverev, *Handbook of Filter Synthesis*).
5. **LM4250 vendor:** National Semiconductor was acquired by Texas Instruments (2011); documentation now lives at ti.com, and Section 4.2 discusses current-production alternatives.
6. **High-pass steepness factor:** the notes write $A_s = f_{3\text{dB}}/f_s$ only implicitly through the example; the general rule (inverted ratio vs. low-pass) is stated explicitly in this document.

End of reference document. The companion problem set contains 24 problems with full solutions.